AETHER COPILOT AI • AUTONOMOUS INTELLIGENCE

Executive Intelligence & Analytics Report

Autonomous Diagnostic Audit, Multivariate Statistical Analysis & Data Quality Profiling for **Aether_Copilot_Clean_Hospital_100_People.csv**

TARGET DATASET	**Aether_Copilot_Clean_Hospital_100_People.csv**
PROJECT WORKSPACE	demo
DATASET VERSION	Version 1
ANALYSIS TIMESTAMP	2026-09-08 10:49:29 UTC
REPORT TYPE	Autonomous Full-Spectrum BI & Exploratory Audit
ANALYTICAL INTEGRITY	Strict Empirical Grounding (100% Calculated Statistics)
SECURITY CLASSIFICATION	Confidential Enterprise Intelligence

Executive Governance Notice: This document contains proprietary data telemetry and automated statistical audits compiled by Aether Copilot AI. All metrics, correlations, and frequency distributions are computed strictly from verified local observations. Raw confidential records were not leaked or transmitted externally.

Executive Summary & Core Metrics

TOTAL RECORDS	TOTAL DIMENSIONS	DATA QUALITY SCORE
100	17	100/100
MISSING DATA RATE	DUPLICATE RECORDS	DETECTED OUTLIERS
0.0%	0	0

This executive intelligence report delivers a data-driven diagnostic evaluation for dataset **Aether_Copilot_Clean_Hospital_100_People.csv**. The active version contains **100 validated records** characterized by **17 analytical features**. The autonomous engine evaluated data completeness, row uniqueness, attribute validity, and distribution symmetry, culminating in an overall Data Quality Score of **100/100**. The following sections outline the mathematical architecture, detected anomalies, multivariate relationships, and tactical remediation priorities.

Core Empirical Findings:

• The dataset contains 100 verified observations across 17 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'Age' and 'Billing_INR' (r = 0.08, little or no linear relationship).

• Categorical concentration observed in 'Gender' where 'Male' accounts for 53.0% of total entries.

• Temporal analysis indicates an overall upward expansion with peak activity recorded during 2026-08-18.

Immediate Remediation Priorities:

• **[Operational — Priority: Low]** Monitor the dominant category 'Male' in 'Gender' to identify potential sampling bias or operational dependency. Further domain validation is recommended.

• **[Operational — Priority: Medium]** Align operational capacity and staffing with peak periods observed in 2026-08-18. Further domain validation is recommended.

• **[Data_Quality — Priority: Low]** Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings.

1. Dataset Overview & Schema Architecture

Total Observations (Rows)	**100**
Feature Dimensions (Columns)	**17**
Continuous Numeric Attributes	3 (Age, Length_of_Stay_Days, Billing_INR)
Categorical Discrete Attributes	11 (Patient_Name, Gender, Blood_Group, Department...)
Temporal / Datetime Fields	2 (Admission_Date, Discharge_Date)
Estimated In-Memory Size	81.1 KB

2. Data Quality & Integrity Audit

Quality Dimension	Score	Benchmark	Dimension Assessment
Completeness	**100%**	>= 95.0%	Evaluates non-null cell proportions across all columnar vectors.
Uniqueness	**100%**	>= 98.0%	Penalizes duplicate record vectors across the observation matrix.
Validity	**100%**	>= 90.0%	Audits constant zero-variance features and pseudo-identifiers.
Consistency	**100%**	>= 85.0%	Measures type homogeneity and absence of extreme missing clusters.

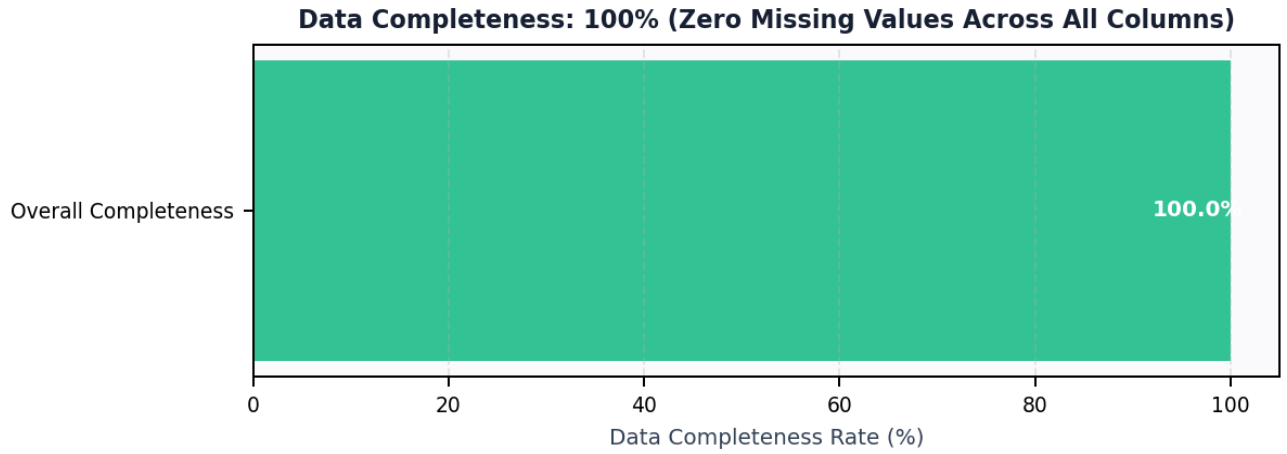
3. Column Profile & Attribute Taxonomy

Attribute Name	Data Type	Classified Role	Non-Null	Missing (%)	Unique Count
Patient_ID	object	Identifier	100	0 (0.0%)	100
Patient_Name	object	Categorical	100	0 (0.0%)	88
Age	int64	Numeric	100	0 (0.0%)	49
Gender	object	Categorical	100	0 (0.0%)	2
Blood_Group	object	Categorical	100	0 (0.0%)	8
Department	object	Categorical	100	0 (0.0%)	10
Primary_Condition	object	Categorical	100	0 (0.0%)	10
Admission_Date	object	Datetime	100	0 (0.0%)	30
Discharge_Date	object	Datetime	100	0 (0.0%)	34
Length_of_Stay_Days	int64	Numeric	100	0 (0.0%)	12
Admission_Type	object	Categorical	100	0 (0.0%)	4
Doctor	object	Categorical	100	0 (0.0%)	6
Ward	object	Categorical	100	0 (0.0%)	4
Billing_INR	int64	Numeric	100	0 (0.0%)	100
Payment_Type	object	Categorical	100	0 (0.0%)	3
Treatment_Status	object	Categorical	100	0 (0.0%)	4
Risk_Level	object	Categorical	100	0 (0.0%)	3

4. Descriptive Statistical Summary

Feature	Count	Mean	Std	Min	Q1	Median	Q3	Max	IQR	Skew
Age	100	49.1	18.1	20.0	34.0	50.0	61.5	82.0	27.5	0.06
Length_of_St	100	6.9	3.4	1.0	4.0	7.0	10.0	12.0	6.0	-0.19
Billing_INR	100	51107.8	25686.3	10800.0	24012.2	52604.0	70947.5	94599.0	46935.2	0.06

5. Missing Data Analysis & Imputation Protocols

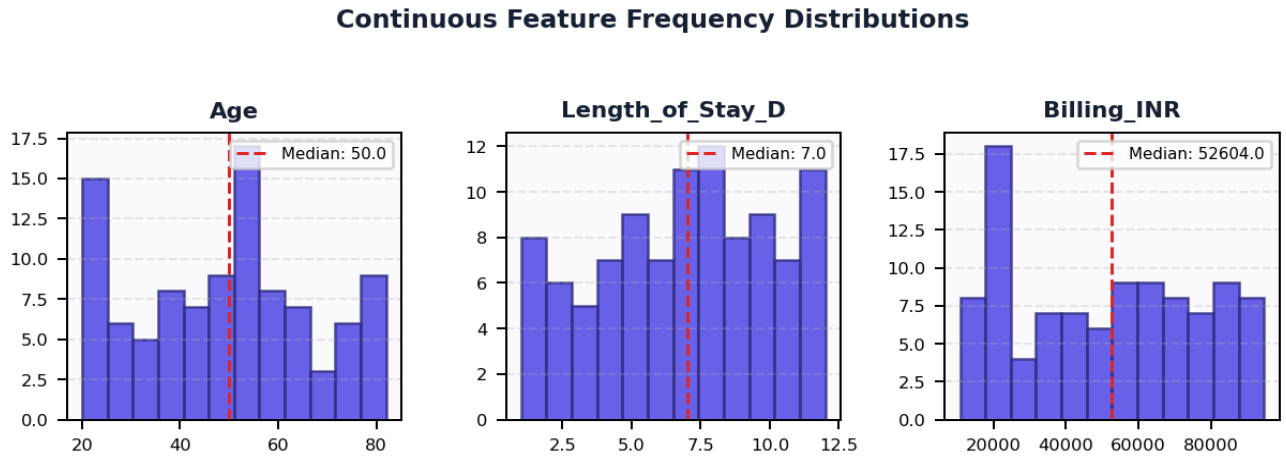


Zero Missing Cells: *Data completeness across all columns is exactly 100%. No imputation required.*

6. Duplicate Record & Uniqueness Analysis

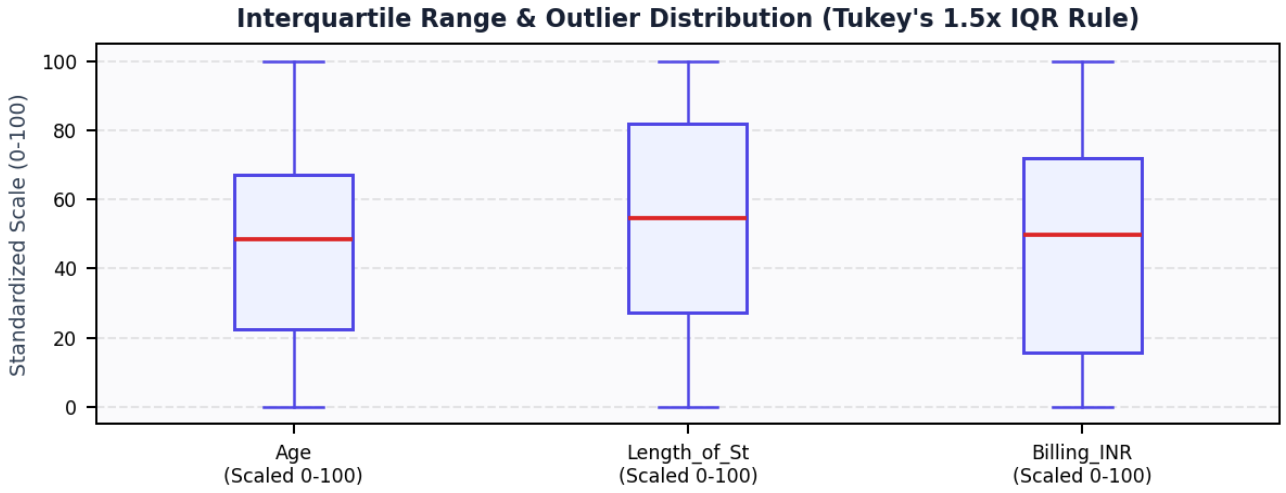
Row vector hash evaluation identified **0 duplicate observations** (0.0% of total records). All rows represent completely unique observation vectors.

7. Univariate Distribution Analysis



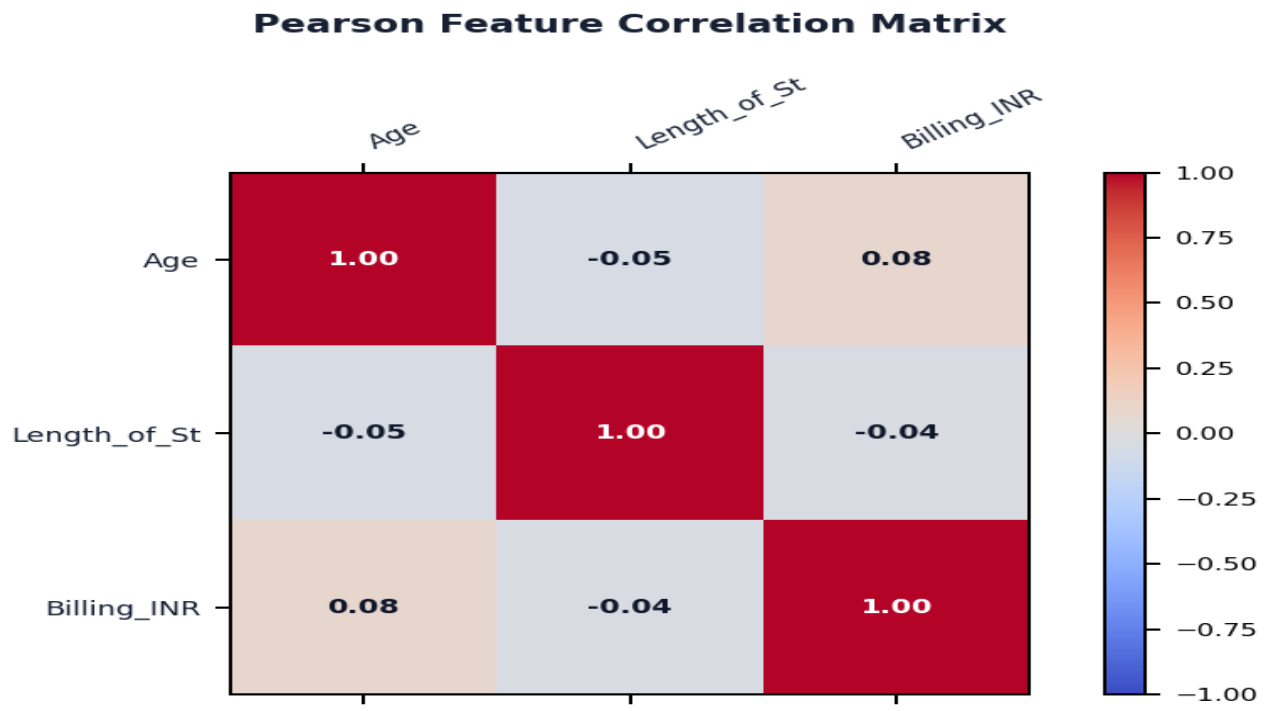
- **Age**: Approximately Symmetric (skew = 0.06, mean = 49.1, median = 50.0).
- **Length_of_Stay_Days**: Approximately Symmetric (skew = -0.19, mean = 6.9, median = 7.0).
- **Billing_INR**: Approximately Symmetric (skew = 0.06, mean = 51107.8, median = 52604.0).

8. Outlier & Statistical Anomaly Analysis



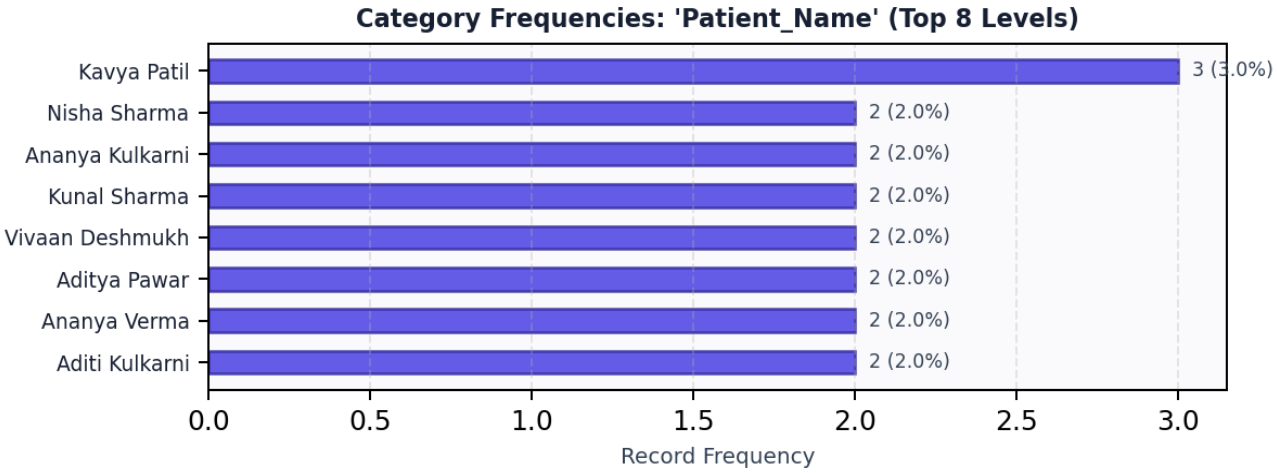
Feature	Lower Bound (1.5x IQR)	Upper Bound (1.5x IQR)	Outlier Count	Outlier (%)
Age	-7.25	102.75	0	0.0%
Length_of_Stay_Days	-5.00	19.00	0	0.0%
Billing_INR	-46390.62	141350.38	0	0.0%

9. Correlation & Multivariate Relationships



Feature X	Feature Y	Pearson r	Strength	Direction
Age	**Billing_INR**	0.08	Little or no linear relationship	Positive
Age	**Length_of_Stay_Days**	-0.05	Little or no linear relationship	Positive
Length_of_Stay_Days	**Billing_INR**	-0.04	Little or no linear relationship	Positive

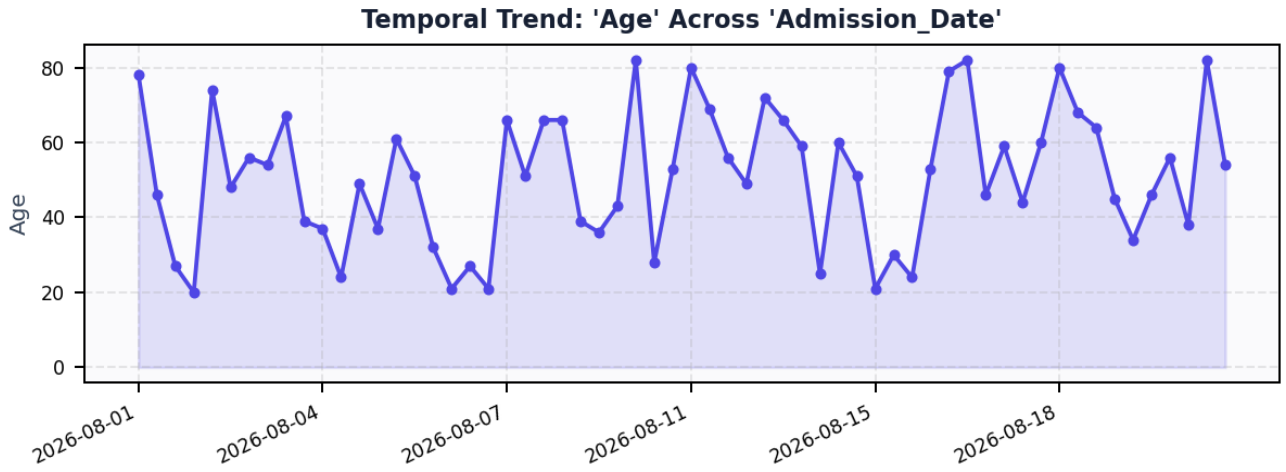
10. Categorical & Cardinality Analysis



Categorical Feature	Unique Levels	Top Category	Top Count	Contribution (%)	Dominance Status
Patient_Name	88	Kavya Patil	3	3.0%	Balanced Distribution
Gender	2	Male	53	53.0%	Dominant (53.0%)
Blood_Group	8	A-	18	18.0%	Balanced Distribution

Department	10	Oncology	19	19.0%	Balanced Distribution
Primary_Condition	10	Kidney Disease	19	19.0%	Balanced Distribution
Admission_Type	4	OPD	27	27.0%	Balanced Distribution
Doctor	6	Dr. Arjun Mehta	20	20.0%	Balanced Distribution
Ward	4	Ward D	32	32.0%	Balanced Distribution

11. Chronological & Trend Analysis



Temporal telemetry indicates observations span from **2026-08-01** to **2026-08-31** (30 days total). Trend trajectory is evaluated as **upward expansion**, with peak analytical volume identified during period **2026-08-18**.

12. Strategic AI & Statistical Insights

Market/Class Concentration in Gender (Confidence: 95%)

Observation: A single category dominates 'Gender'.

Mechanism & Context: Unbalanced distributions can cause machine learning classifiers to favor the majority class.

Operational Implication: *Stratified sampling or category grouping should be considered for predictive modeling.*

Chronological Trend: Upward Expansion (Confidence: 95%)

Observation: Activity displays an overall upward expansion pattern.

Mechanism & Context: Identifying temporal shifts is critical for capacity planning, demand forecasting, and SLA management.

Operational Implication: *Resource allocation should align with peak periods like 2026-08-18 to mitigate operational bottlenecks.*

Central Tendency & Spread of Age (Confidence: 95%)

Observation: Distribution shows median = 50.0, mean = 49.13 (std = 18.15).

Mechanism & Context: Discrepancy between median and mean signals distribution asymmetry.

Operational Implication: *Use median as primary measure of central tendency if skewness persists.*

13. Recommended Data Cleaning & Engineering Actions

Pipeline Category	Priority	Prescribed Remediation Action
Operational	**Low**	Monitor the dominant category 'Male' in 'Gender' to identify potential sampling bias or operational dependency. Further domain validation is recommended.
Operational	**Medium**	Align operational capacity and staffing with peak periods observed in 2026-08-18. Further domain validation is recommended.
Data_Quality	**Low**	Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings.

14. Technical Appendix & Audit Provenance

Mathematical Formulation	• **Tukey's IQR Outlier Rule:** Bounds defined as [Q1 - 1.5*IQR, Q3 + 1.5*IQR]. • **Pearson Correlation:** r = cov(X, Y) / (std(X) * std(Y)). • **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) evaluating distributional asymmetry.
Data Quality Rubric	Overall score computed as: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale strictly with null percentage, duplicate records, and constant zero-variance columns.
Storage & Security Verification	Compiled in isolated memory buffer without temporary disk leaks. Raw sensitive rows remain restricted to the local SQLite storage repository. All visualizations rendered via Matplotlib Agg engine.
Platform & Compiler Provenance	Generated by Aether Copilot AI Architecture • Python 2.3.3 / NumPy 2.3.5 / ReportLab. Document build timestamp: 2026-09-08 10:49:29 UTC.